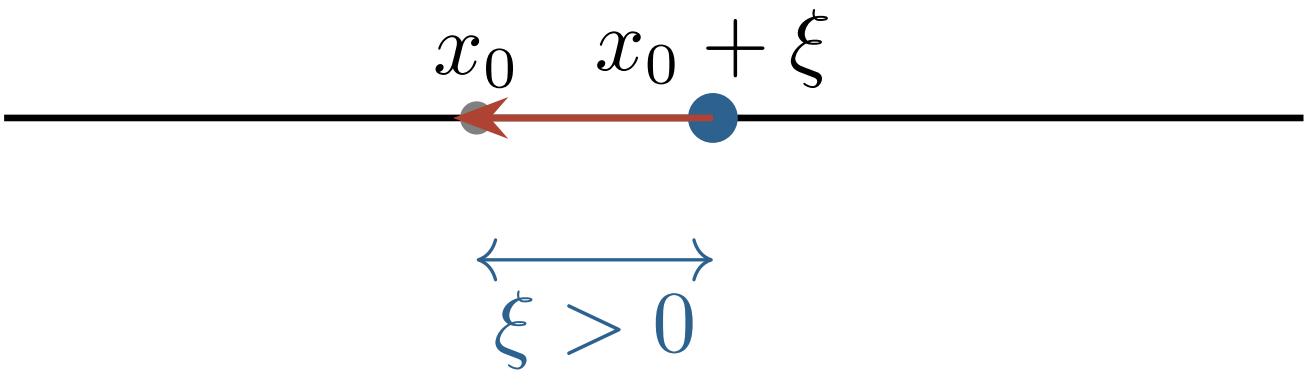
$$F \simeq F'(x_0)\xi < 0$$



$$m\ddot{\xi} + \kappa\xi = 0$$

$$\kappa = K(a^2 - 2x_0^2)/(a^2 + x_0^2)^{5/2} > 0$$